

MATHEMATICAL FOUNDATIONS FOR SCIENTIFIC THINKING

Class I

The Map of Mathematics: A Survey of the Essential Disciplines

Course Overview & Curriculum Plan | 2025–2026

Beryl Meir

Department of Electrical Engineering | Bar-Ilan University | Ramat Gan, Israel

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$e^{i\pi} + 1 = 0$$

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx$$

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

COURSE MAP

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PREFACE

Purpose & Pedagogical Philosophy

Who This Course Is For

This curriculum is designed for technically mature people — graduate students, researchers, engineers, and scientists — who are highly intelligent but whose mathematical foundations have the shape of their specific training rather than genuine structural breadth. The goal is not to survey every branch of mathematics, but to build the reusable mathematical and computational machinery underlying all serious technical disciplines.

Structure & Approach

The course is organised not by traditional academic subjects, which often reflect institutional history rather than cognitive architecture, but by *modes of technical thinking*. Each unit treats one fundamental way of understanding the world: through change, through structure, through uncertainty, through decomposition. Subjects such as Fourier analysis or Laplace transforms appear not as isolated topics but as natural tools within a broader narrative.

Depth vs. Coverage

We do not aim for complete mastery of every topic. For each concept the course provides rigorous intuition, one or two carefully chosen examples, connections to adjacent areas, and a clear indication of where further study begins. The deliberate goal is to bring the student to the threshold of independent learning.

A Note on Notation

Mathematical notation is introduced from the first session not as an obstacle but as a language. Fluency in notation is treated as an explicit learning objective, because without it, reading technical literature remains permanently effortful.

*“The map is not the territory — but a good map makes exploration possible.
This course is a map.”*

UNIT I

Mathematical Language & Logical Foundations

Learning Goal *Become fluent in the grammar of mathematics: reading, writing, and reasoning with precision.*

Every technical discipline is written in a shared formal language. This unit introduces that language from first principles — not as a set of rules to memorise, but as a system of thought. We establish what it means to prove something, how mathematical objects are defined and related, and how to reason from hypotheses to conclusions without ambiguity.

Sets & Set Operations

$$A \cup B, \quad A \cap B, \quad A \setminus B, \quad A \subseteq B$$

The foundation of all mathematical structure. A set is a collection of objects; operations on sets encode union, intersection, and containment. Every later concept — functions, spaces, algebras — is built on sets.

Functions & Mappings

$$f : X \rightarrow Y, \quad f \circ g, \quad f^{-1}$$

Functions are the central objects of mathematics: they encode relationships, transformations, and processes. Injectivity, surjectivity, and bijectivity are the key structural properties.

Logic & Quantifiers

$$\forall x \in X, \exists y \in Y \text{ s.t. } P(x, y)$$

Formal logic provides the language of rigorous argument. Universal and existential quantifiers structure every theorem and proof. Learning to negate quantified statements is an essential technical skill.

Proof Techniques

$$P \Rightarrow Q \iff \neg Q \Rightarrow \neg P$$

Direct proof, proof by contradiction, proof by contrapositive, and mathematical induction. These are the four primary modes of mathematical argument, each suited to different problem structures.

Sequences & Limits

$$\lim_{n \rightarrow \infty} a_n = L \iff \forall \varepsilon > 0, \exists N : n > N \Rightarrow |a_n - L| < \varepsilon$$

The ε - N definition makes rigorous the intuitive notion of “approaching a value.” Limits underlie continuity, derivatives, integrals, and series — essentially all of analysis.

UNIT II

Algebraic & Symbolic Fluency

Learning Goal *Make symbolic manipulation second nature, eliminating cognitive overhead from technical reading.*

A large fraction of technical difficulty is not conceptual but notational: expressions that are opaque because the reader is not fluent in the algebraic language. This unit focuses on building automatic fluency with the core symbolic toolkit — not as a drill exercise, but by understanding why each family of operations exists and what it encodes.

Exponentials & Logarithms

$$e^{a+b} = e^a e^b, \quad \ln(xy) = \ln x + \ln y$$

Exponentials encode multiplicative growth; logarithms invert them and convert products to sums. They appear in growth models, information theory, signal processing, and thermodynamics.

Trigonometry & the Unit Circle

$$\sin^2 \theta + \cos^2 \theta = 1, \quad e^{i\theta} = \cos \theta + i \sin \theta$$

Trigonometric functions parameterise circular motion and encode oscillation. Euler's formula unifies them with the exponential, which is the gateway to Fourier analysis.

Complex Numbers

$$z = a + bi, \quad |z| = \sqrt{a^2 + b^2}, \quad \bar{z} = a - bi$$

Complex numbers complete the algebraic closure of the reals and are indispensable in signal processing, quantum mechanics, and control theory. The complex plane gives geometric intuition for algebraic operations.

Summation & Product Notation

$$\sum_{k=1}^n f(k), \quad \prod_{k=1}^n f(k)$$

Compact notation for aggregate operations. Facility with index manipulation, telescoping sums, and geometric series is required across probability, algorithms, and analysis.

Asymptotic Notation

$$f(n) = O(g(n)) \iff \exists C, n_0 : |f(n)| \leq C|g(n)| \quad \forall n > n_0$$

Big-O and related notations describe behaviour at scale without committing to exact constants. Indispensable for algorithm analysis, approximation theory, and perturbation methods.

UNIT III

Change & Accumulation — Single Variable

Learning Goal *Understand local behaviour, rates of change, and accumulated quantities as the language of natural law.*

Calculus is the mathematics of change. Almost every equation governing a physical, biological, economic, or computational system is written in the language of derivatives and integrals. This unit builds that language from the ground up: what a derivative really measures, what an integral really accumulates, and how the Fundamental Theorem ties them together.

The Derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

The derivative measures the instantaneous rate of change. Geometrically it is the slope of the tangent line; physically it is velocity, current, or any rate of flow. Chain rule, product rule, and implicit differentiation are the core manipulation tools.

The Integral

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

The integral accumulates infinitesimal contributions — area, mass, charge, probability. The Riemann definition grounds the concept; the Fundamental Theorem connects it to differentiation.

The Fundamental Theorem of Calculus

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

The central result of calculus: differentiation and integration are inverse operations. This theorem makes symbolic integration tractable and justifies the antiderivative approach.

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

Every smooth function can be locally approximated by a polynomial. Taylor series are the fundamental tool for approximation, numerical methods, and understanding local behaviour — and they reappear throughout physics and engineering.

Convergence of Series

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

When do infinite sums converge? Ratio test, comparison test, and integral test provide the core criteria. Convergence questions arise wherever infinite processes appear: Fourier series, probability generating functions, and quantum mechanics.

UNIT IV

Multivariable Calculus & Optimization

Learning Goal *Extend single-variable intuition to functions of many variables — the natural setting of real systems.*

Real systems depend on many variables simultaneously. Temperature depends on position in space; the cost of a process depends on dozens of parameters; a neural network output depends on millions of weights. Multivariable calculus provides the tools to reason about change, approximation, and optimization in these high-dimensional settings.

Partial Derivatives & the Gradient

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

The gradient generalises the derivative to multiple dimensions: it points in the direction of steepest ascent and has magnitude equal to the rate of increase in that direction. It is the engine of gradient descent and variational methods.

The Jacobian

$$J_{ij} = \frac{\partial f_i}{\partial x_j}$$

For vector-valued functions, the Jacobian matrix collects all partial derivatives. It encodes the local linear approximation of the map and appears in coordinate changes, robotics, and sensitivity analysis.

The Hessian & Second-Order Behaviour

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

The Hessian matrix captures curvature. A positive-definite Hessian implies a local minimum; a negative-definite Hessian implies a local maximum. Central to optimisation and the analysis of critical points.

Multiple Integrals & Change of Variables

$$\iint_D f \, dA = \int \int f(r, \theta) \, r \, dr \, d\theta$$

Integration over regions in two or three dimensions. Change of variables (via the Jacobian determinant) allows transformation between coordinate systems — Cartesian, polar, spherical — essential for physics and probability.

Constrained Optimisation & Lagrange Multipliers

$$\nabla f = \lambda \nabla g \quad \text{at a constrained extremum}$$

Most real optimisation problems have constraints. Lagrange multipliers convert a constrained problem into an unconstrained system. The method underlies everything from thermodynamic equilibrium to portfolio optimisation.

UNIT V

Linear Structure

Learning Goal *Master the single most important mathematical framework in modern science and computation.*

Linear algebra is not merely a tool — it is a way of seeing. Data science, quantum mechanics, signal processing, structural engineering, and machine learning all reduce, at their core, to questions about linear maps, vector spaces, and matrix decompositions. This unit builds genuine structural understanding, not just computational technique.

Vector Spaces & Subspaces

$$V = \text{span}\{v_1, \dots, v_k\}, \quad \dim V = k$$

A vector space is any collection of objects that can be added and scaled consistently. The key concepts are spanning sets, linear independence, bases, and dimension — which measure the intrinsic degrees of freedom.

Linear Maps & Matrices

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

A linear map preserves the structure of the vector space. Matrices represent linear maps in coordinates. Matrix multiplication encodes composition of maps, not just a formal operation.

Eigenvalues & Eigenvectors

$$\mathbf{A} \mathbf{v} = \lambda \mathbf{v}$$

Eigenvectors are the natural directions of a linear map — unchanged in direction under the transformation. Eigenvalues measure the associated scaling. Spectral decomposition reveals the intrinsic geometry of a matrix.

Orthogonality & Least Squares

$$\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 \implies \mathbf{A}^\top \mathbf{A} \mathbf{x} = \mathbf{A}^\top \mathbf{b}$$

Orthogonality is the key to projections and approximation. Least squares finds the best linear fit to overdetermined data — the foundation of regression, filtering, and signal reconstruction.

Singular Value Decomposition

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$$

The SVD is the universal matrix decomposition. It reveals the rank, the principal directions of action, and the best low-rank approximations of any matrix. It underlies PCA, image compression, and latent semantic analysis.

UNIT VI

Discrete & Algorithmic Thinking

Learning Goal *Develop the combinatorial and computational mode of reasoning that governs discrete and finite systems.*

Not all mathematics is continuous. Counting, graph structures, recursion, and algorithmic complexity form a distinct and essential mode of technical thought. Discrete mathematics is the native language of computer science, combinatorics, cryptography, and network theory — and it provides indispensable tools for probability and algorithm design.

Combinatorics & Counting

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

How many ways can objects be arranged, selected, or distributed? Permutations, combinations, and the binomial theorem answer these questions. Counting arguments underlie probability, algorithm analysis, and coding theory.

Graph Theory

$$G = (V, E), \quad \deg(v) = |\{u : (u, v) \in E\}|$$

Graphs encode relationships: molecules, networks, dependencies, social structures, circuits. Key concepts are paths, connectivity, trees, and colouring — which reappear in scheduling, optimisation, and computational complexity.

Recursion & Mathematical Induction

$$P(1) \text{ true}, \quad P(k) \Rightarrow P(k+1) \implies P(n) \text{ true } \forall n$$

Induction is the proof technique native to discrete structures. Recursion is the corresponding computational pattern. Together they underlie the analysis of algorithms and the structure of data.

Algorithmic Complexity

$$T(n) = O(n \log n), \quad T(n) = \Omega(n^2)$$

How does the cost of computation scale with input size? Big-O analysis measures efficiency. The P vs. NP distinction is the deepest open problem in theoretical computer science and has profound practical implications.

Basic Algorithmic Paradigms

$$T(n) = 2T(n/2) + O(n) \implies T(n) = O(n \log n)$$

Divide-and-conquer, dynamic programming, and greedy algorithms are the three foundational design strategies. Understanding them enables recognition of structure in novel problems, not just memorisation of known solutions.

UNIT VII

Uncertainty & Information

Learning Goal *Become capable of reasoning rigorously under uncertainty and quantifying the content of data.*

The world is stochastic. Measurements have noise; models are approximate; outcomes are uncertain. Probability provides the mathematical framework for quantifying uncertainty, and statistics provides the tools for drawing defensible conclusions from data. Information theory reveals a deep connection between probability and the compressibility of messages.

Probability Spaces & Axioms

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Probability is a measure on a space of outcomes. The Kolmogorov axioms make this precise. Events, sample spaces, and sigma-algebras are the foundational objects from which everything else is built.

Random Variables & Distributions

$$\mathbb{E}[X] = \int x f(x) dx, \quad \text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

A random variable maps outcomes to numbers; its distribution describes the probability of each value. Key distributions — Gaussian, Poisson, Bernoulli, exponential — arise repeatedly across science and engineering.

Conditional Probability & Bayes' Theorem

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Bayes' theorem is the fundamental rule for updating beliefs in light of evidence. It underlies Bayesian statistics, machine learning, Kalman filtering, and medical diagnosis.

Limit Theorems

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

The Law of Large Numbers and the Central Limit Theorem explain why averages converge and why Gaussian distributions are so prevalent. They are the theoretical bedrock of statistical inference.

Entropy & Information Theory

$$H(X) = - \sum_x p(x) \log_2 p(x)$$

Shannon entropy measures the average information content of a random variable — equivalently, the irreducible randomness in a source. From it flows data compression, channel capacity, and the thermodynamic interpretation of information.

UNIT VIII

Dynamics & Systems

Learning Goal *Understand how systems evolve in time and how feedback shapes their long-term behaviour.*

Most systems of interest — physical, biological, chemical, economic — are not static but dynamic: they evolve according to rules expressed as differential equations. This unit builds the tools to analyse such systems: finding solutions, understanding stability, and seeing the qualitative behaviour of trajectories without necessarily solving equations explicitly.

First-Order ODEs

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

The simplest dynamical systems. Separation of variables and integrating factors are the primary solution techniques. Exponential growth and decay, Newton's law of cooling, and RC circuits are canonical examples.

Systems of ODEs & Phase Space

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad \mathbf{x}(t) = e^{At}\mathbf{x}_0$$

Vector differential equations describe multi-component systems. Phase portraits — plots of trajectories in state space — give qualitative insight into long-term behaviour without explicit solutions.

Stability & Linearisation

$$\lambda = \alpha \pm \beta i : \quad \alpha < 0 \Rightarrow \text{stable}, \quad \alpha > 0 \Rightarrow \text{unstable}$$

Near a fixed point, a nonlinear system behaves like its linearisation. The eigenvalues of the Jacobian determine stability. This principle underlies perturbation theory across physics, biology, and control engineering.

The Laplace Transform

$$\mathcal{L}\{f\}(s) = \int_0^\infty f(t) e^{-st} dt$$

The Laplace transform converts differential equations into algebraic equations, making them far easier to solve and manipulate. It is the standard tool for linear time-invariant systems analysis in engineering.

Feedback & Control Intuition

$$G_{cl}(s) = \frac{G(s)}{1 + G(s)H(s)}$$

Feedback is the mechanism by which a system regulates itself. Negative feedback promotes stability; positive feedback promotes growth or bifurcation. Control theory formalises the design of feedback to achieve desired system behaviour.

UNIT IX

Frequency & Decomposition

Learning Goal *Learn to decompose signals and functions into their constituent frequencies — a universal analytical strategy.*

One of the most powerful ideas in all of mathematics and physics is that complicated functions can be expressed as superpositions of simple oscillations. This insight — due to Fourier — has transformed physics, engineering, and data science. Frequency-domain thinking is a distinct mode of analysis that complements the time-domain perspective and makes many hard problems tractable.

Fourier Series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Periodic functions can be expressed as infinite sums of sinusoids. The Fourier coefficients measure how much of each frequency is present. Fourier series are the foundation of acoustics, image compression, and PDE solving.

The Fourier Transform

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx$$

The continuous Fourier transform extends Fourier series to non-periodic functions. It decomposes an arbitrary signal into a continuous spectrum of frequencies. The Uncertainty Principle is a direct consequence of its properties.

Convolution

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

Convolution measures the overlap of two functions as one slides over the other. It encodes filtering, blurring, and response to input. The convolution theorem — that convolution becomes multiplication under the Fourier transform — is fundamental to signal processing.

Filtering & the Frequency Domain

$$\hat{y}(\xi) = H(\xi) \hat{x}(\xi)$$

A filter acts multiplicatively in the frequency domain: it attenuates certain frequencies and passes others. Low-pass, high-pass, and band-pass filters are ubiquitous in audio, image processing, communications, and neuroscience.

Discrete Fourier Transform & the FFT

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n / N}$$

The DFT adapts Fourier analysis to sampled signals. The Fast Fourier Transform computes it in $O(N \log N)$ operations, making spectral analysis computationally feasible. It is one of the most important algorithms in applied mathematics.

UNIT X

Continuous Fields & PDE Thinking

Learning Goal Understand how fields — temperature, pressure, concentration, displacement — evolve across space and time.

When a quantity depends on both space and time, ordinary differential equations are no longer sufficient. Partial differential equations govern the evolution of continuous fields: heat diffusion, wave propagation, fluid flow, quantum states. This unit develops intuition for the major PDE types and the techniques for analysing and solving them.

Classification of PDEs

$$u_{tt} - c^2 u_{xx} = 0 \text{ (hyperbolic),} \quad u_t - k u_{xx} = 0 \text{ (parabolic)}$$

Elliptic, parabolic, and hyperbolic PDEs describe fundamentally different physical phenomena: equilibria, diffusion, and wave propagation respectively. Classification determines the correct boundary conditions and solution methods.

The Heat Equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

The paradigm of diffusion. It describes heat conduction, Brownian motion, and concentration gradients. Smooth initial data becomes smoother over time — a qualitative signature of parabolic equations.

The Wave Equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

The paradigm of propagation. D'Alembert's solution shows that disturbances travel at speed c . Wave equations govern sound, light, seismic waves, and quantum probability amplitudes.

Separation of Variables

$$u(x, t) = X(x) T(t) \implies \frac{T'}{T} = \frac{X''}{X} = -\lambda$$

The primary analytical technique for linear PDEs with simple geometry. It reduces a PDE to a pair of ODEs connected by a separation constant. Combined with Fourier series, it yields explicit solutions on bounded domains.

Boundary Conditions & Well-Posedness

$$u(0, t) = u(L, t) = 0 \text{ (Dirichlet)}$$

The physical domain imposes constraints on solutions. Dirichlet (value), Neumann (flux), and Robin conditions correspond to fixed temperature, insulated boundary, and convective exchange respectively. A well-posed problem has a unique, stable solution.

UNIT XI

Computation & Numerical Methods

Learning Goal *Bridge the gap between mathematical analysis and actual numerical computation.*

Mathematics does not always yield closed-form answers. In practice, most differential equations, integrals, and optimisation problems must be solved numerically. This unit builds an understanding of how numerical algorithms work, where they fail, and how to reason about the reliability of computed results — skills without which mathematical knowledge cannot be applied to real systems.

Floating-Point Arithmetic & Numerical Error

$$\text{fl}(x) = x(1 + \delta), \quad |\delta| \leq \epsilon_{\text{mach}}$$

Real numbers are approximated in finite precision. Machine epsilon, round-off error, and catastrophic cancellation determine the limits of numerical computation. Every numerical result carries an implicit error budget.

Root Finding

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (\text{Newton's method})$$

Finding zeros of functions is a fundamental computational task: equilibria, intersections, and eigenvalues are all root problems. Bisection is safe but slow; Newton's method is fast but requires a good initial guess.

Numerical Integration

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k)$$

When antiderivatives are unavailable, quadrature rules (trapezoid, Simpson, Gaussian) approximate integrals by weighted sums at sample points. Error analysis determines which rule is appropriate for a given function.

Numerical Linear Algebra

$$A = LU, \quad A = QR, \quad A = U\Sigma V^T$$

Solving linear systems and computing eigenvalues numerically requires factorisation methods. LU, QR, and SVD decompositions are the workhorses of scientific computing. Condition number measures how sensitive a problem is to perturbation.

Numerical ODEs & Simulation

$$y_{n+1} = y_n + h f(t_n, y_n) \quad (\text{Euler's method})$$

Differential equations are integrated numerically using step-by-step methods. Euler's method illustrates the idea; Runge-Kutta and adaptive step-size methods achieve the accuracy required in practice. Stability dictates the maximum step size.